

■ B.1.6 Numbers

Integer	integer
Real	real (floating point) number
Rational	rational number
Complex	complex number

Basic types of numbers.

All numbers in *Mathematica* can contain any number of digits. Floating point numbers can have any exponent up to some maximum absolute value which depends on the implementation.

Numbers with sufficiently few digits are usually implemented as “machine numbers”. On most computer systems, this is true for integers less than $2^{31} \simeq 2 \times 10^9$ and floating point numbers with fewer than 16 digits. Operations on machine numbers are usually much more efficient than operations on arbitrary-precision numbers.

Floating point numbers input with a small number of digits are treated with the precision of machine numbers.

Numbers can be entered in any base from 2 to 36, using the notation $\text{base}^{\text{digits}}$. *base* is given in decimal. For bases larger than 10, additional digits are chosen from the letters a–z or A–Z. Upper and lower case letters are equivalent for these purposes. Floating point numbers can be specified by including . in the *digits* sequence.